Framingham Longitudinal Stroke Risk Profile

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Introduction

This project investigates stroke risk in the Framingham Heart Study, a long-term prospective cohort that has followed adults in Framingham, Massachusetts for cardiovascular outcomes since 1948. This analysis uses a subset of 4,434 participants with baseline measurements of cardiovascular risk factors and stroke outcomes. To ensure that covariates reflect clinically relevant exposure windows, this project focuses on the first 10 years of follow-up.

The primary objective is to identify baseline predictors of time to first stroke, analyzed separately for males and females. A secondary aim is to estimate 10 year stroke probabilities for several clinically relevant risk profiles, defined by age, systolic blood pressure (BP), and diabetes and smoking status. Because systolic BP and diabetes were measured at three exam periods, an additional aim is to describe how these risk factors change over time and to discuss the implications for analyses that rely only on baseline values.

Methods

All analyses were conducted in R v4.5.2 (2025-10-31 ucrt). The analytic dataset included 4,434 participants with baseline measurements of cardiovascular risk factors and adjudicated stroke outcomes. Individuals with evidence of stroke at baseline were excluded. Follow-

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up time was defined using the TIMESTRK variable, representing time to first stroke or censoring. Followup was restricted to the first 10 years. Individuals who experienced a stroke within 10 years were coded as events. Those that remained stroke free were censored at 10 years or at their last known followup time.

The primary outcome was time to first stroke (atherothrombotic infarction or cerebral hemorrhage) within 10 years of baseline. Participants who died before experiencing a stroke were censored at time of death. Other non-stroke cardiovascular events were not treated as competing risks and did not alter censoring. Baseline covariates include age, systolic BP, diabetes, smoking status, BMI, cholesterol, coronary heart disease, and blood pressure medication use. Covariates were analyzed and recorded at baseline. No time varying covariates were included in the primary survival models. Complete case analysis was used for the final Cox models, with participants excluded only if missing data occurred in variables included in the final model. Analyses were stratified by sex, with separate models fit for males and females.

Kaplan–Meier curves and log-rank tests were used to estimate stroke-free survival, and log-rank tests compared survival across categories of diabetes, systolic BP, and smoking by sex. Cox proportional hazards models were used to identify baseline predictors of stroke. Age, systolic blood pressure, and diabetes were included *a priori* based on clinical relevance. Additional covariates were evaluated using backwards selection based on AIC. Proportional hazards assumptions were assessed using Schoenfeld residuals and log-minus-log survival curves. For each sex, 10 year predicted stroke probabilities were estimated from the final Cox model for five predefined risk profiles at ages 40, 50, and 60: 1. No baseline risk factors,

2. High systolic blood pressure ($\geq$ 160), 3. Diabetes, 4. Both high systolic blood pressure and diabetes, 5. One additional profile chosen based on model results (current smoker).

Finally, systolic BP and diabetes status were summarized across the three exam periods to assess how much these risk factors changed over time and to consider if time-varying covariate models may be appropriate for future analyses.

Results

Table 1 summarizes baseline characteristics for the analytic dataset. Males and females were similar in age and diabetes prevalence, though females had slightly higher systolic BP and total cholesterol. Smoking was substantially more common in males (60.5% vs. 40.5%).

Kaplan-Meier curves demonstrated differences in stroke-free survival across several baseline risk factors (**Figure 1**). Diabetes and high systolic BP showed the strongest separation of survival curves, while smoking had minimal impact. Among males, the difference in diabetes was pronounced (X^2 = 28.5, p « 0.001), with a more modest effect in females (X^2 = 6.9, p = 0.009). High systolic BP ($\geq$ 160 mmHg) showed the largest differences overall, with highly significant log-rank tests for both males (X^2 = 68.4, p « 0.001) and females (X^2 = 47.7, p « 0.001). In contrast, smoking status was not associated with stroke-free survival in either males or females (p = 0.5; 0.4).

Separate Cox proportional hazards models were fit for males and females. Age, systolic BP, and diabetes were included *a priori*, and backward selection supported retaining smoking status for males. To aid in comparability across sexes, smoking was also included in the female model.

For males, age, systolic BP, diabetes, and smoking were all significant predictors of stroke. For every one year increase in age, the hazard of stroke increased by 6.6% on average (HR = 1.07, 95% CI: 1.03, 1.11). Each 1 mmHg increase in systolic BP was associated with a 3.4% increase in hazard (HR = 1.03; 95% CI: 1.02, 1.05). Diabetes was a strong predictor, with diabetic men experiencing nearly five times the hazard of non-diabetic men (HR = 4.81, 95% CI: 2.10, 11.01). Smoking was also associated with elevated risk (HR = 1.92, 95% CI: 1.04, 3.57). The proportional hazards assumption was generally met. Diabetes showed mild deviation (p = 0.048), but the global test was non-significant (p = 0.23), validated by visual inspection (**Figure 2**).

For females, age and systolic BP were strong predictors of stroke. Each additional year of age increased the hazard by 7.8% (HR = 1.08, 95% CI: 1.04, 1.12), and each 1 mmHg increase in systolic BP increased the hazard by 2.6% (HR = 1.03, 95% CI: 1.02, 1.04). Diabetes and smoking were not statistically significant predictors in females (p = 0.25; 0.21). Proportional hazards assumptions were all met with a global p - 0.96 (**Figure 2**).

Predicted 10 year stroke-free survival was estimated for five risk profiles (**Table 2**). Across all profiles, stroke risk increased with age and was consistently higher in males than females (**Figure 3**). High systolic BP produced the largest increase in predicted risk for both sexes. For example, among males aged 60, stroke-free survival declined from 98.9% for no baseline risk factors to 95.9% with high BP and 81.6% with high BP and diabetes combined. Diabetes had a stronger impact in males than females, consistent with the Cox model results.

To assess whether time-varying covariate models might be warrented in future research, systolic BP and diabetes were summarized across the three exam periods (**Figure 4**). Mean systolic BP increased steadily over time for both sexes, rising approximately 7 mmHg from period 1 to 3. Diabetes prevalence also increased from roughly 3% at period 1 to 8% at period 3.

Discussion

This paper evaluated baseline predictors of 10 year stroke risk in the Framingham Heart Study and generated clinically interpretable risk estimates for several common risk profiles. Across all analyses, systolic BP emerged as the strongest predictor of stroke, with diabetes contributing additional risk (particularly among males). Predicted 10 year stroke probabilities highlight the impact of elevated BP and the compounded risk when it occurs with diabetes. Risk increased steadily with age and was consistently higher in males, though sex differences were modest overall.

Although the primary analysis relied on baseline covariates, systolic BP and diabetes both changed meaningfully across exam periods, with rising blood pressure and increasing diabetes prevalence. These shifts suggest that time-varying covariate models may be more accurate in future work. This is amongst other limitations that should be considered. Complete-case analysis may introduce bias related to unmeasured factors. Competing risks such as death were not explicitly modeled, which may overestimate stroke risk. Additionally, the Framingham cohort may limit generalizability to more diverse groups. Dispite these constraints, there is clear evidence of baseline predictors of stroke and clinically relevant 10 year risk estimates.

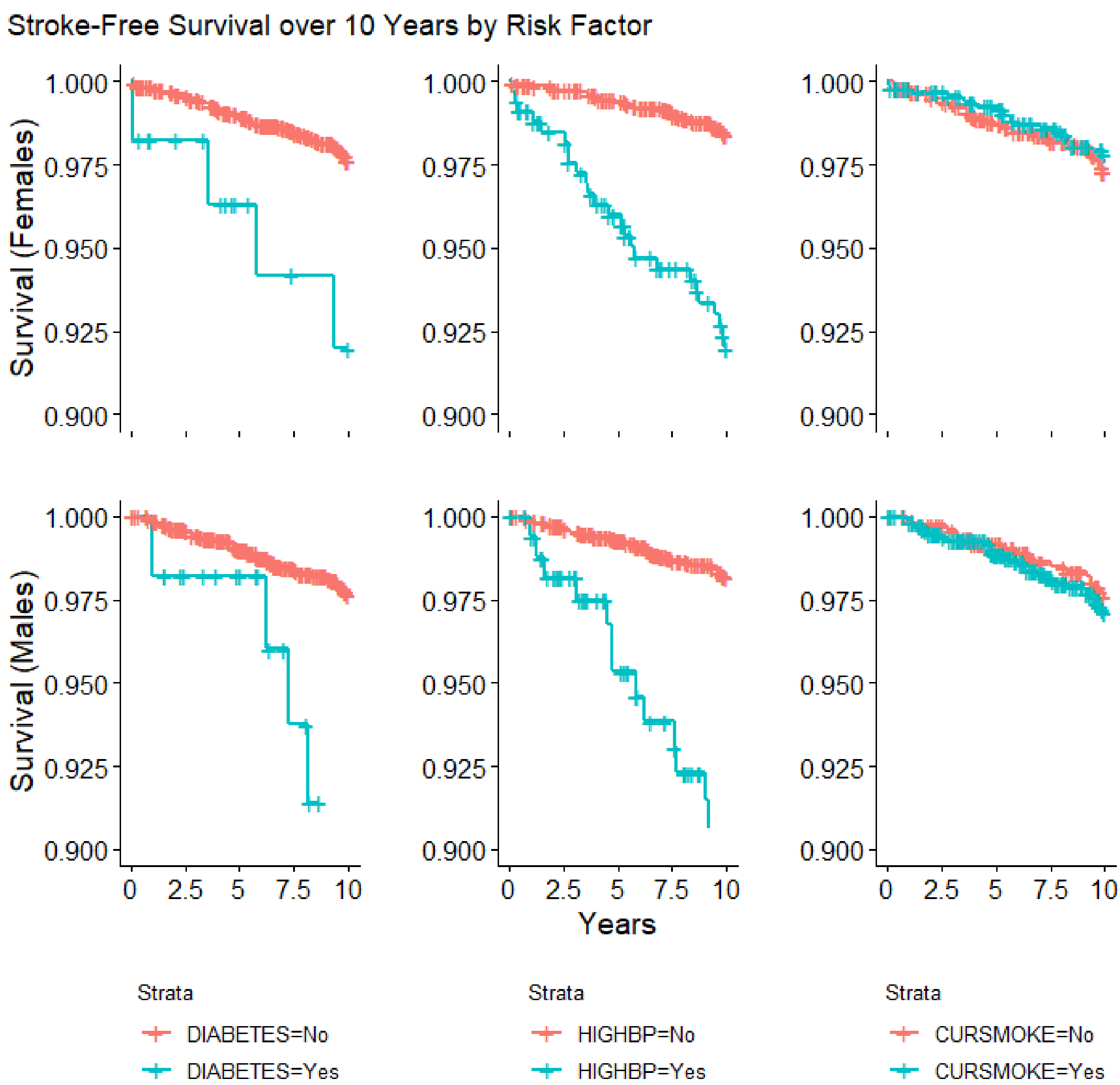
Appendix

Code Directory: https://github.com/eweible/BIOS6624.git

File name: P3/Code/P3_Report.Rmd

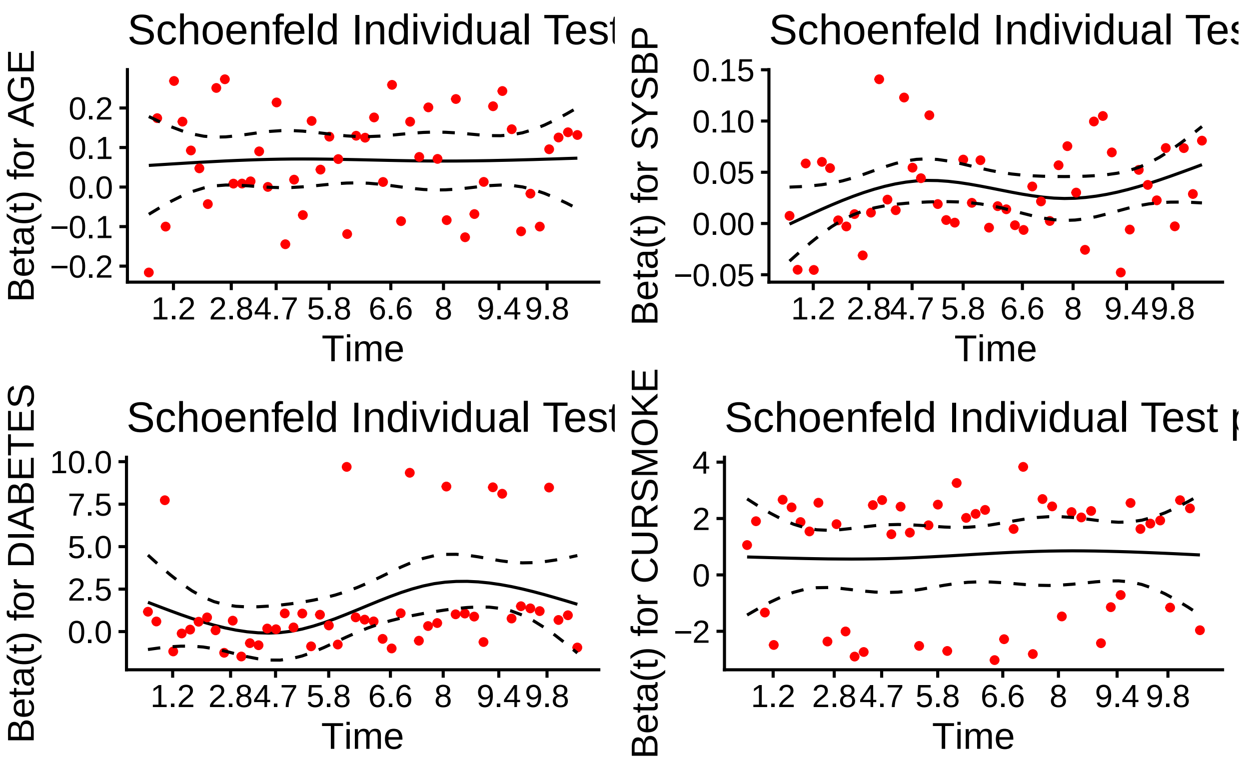
Data Directory: C:/Users/weiblee/OneDrive/Documents/Anschutz/Spring 2026/6624_Advanced/Weible_Advanced/P3/DataRaw

	Male (N=1930)	Female (N=2472)	Overall (N=4402)
AGE			
Mean (SD)	49.7 (8.71)	50.0 (8.63)	49.9 (8.66)
Median [Min, Max]	49.0 [33.0, 69.0]	49.0 [32.0, 70.0]	49.0 [32.0, 70.0]
SYSBP			
Mean (SD)	132 (19.4)	134 (24.3)	133 (22.3)
Median [Min, Max]	129 [83.5, 235]	129 [83.5, 295]	129 [83.5, 295]
DIABETES			
No	1871 (96.9%)	2412 (97.6%)	4283 (97.3%)
Yes	59 (3.1%)	60 (2.4%)	119 (2.7%)
PREVCHD			
No	1810 (93.8%)	2405 (97.3%)	4215 (95.8%)
Yes	120 (6.2%)	67 (2.7%)	187 (4.2%)
BPMEDS			
No	1868 (96.8%)	2338 (94.6%)	4206 (95.5%)
Yes	40 (2.1%)	96 (3.9%)	136 (3.1%)
Missing	22 (1.1%)	38 (1.5%)	60 (1.4%)
CURSMOKE			
No	762 (39.5%)	1471 (59.5%)	2233 (50.7%)
Yes	1168 (60.5%)	1001 (40.5%)	2169 (49.3%)
TOTCHOL			
Mean (SD)	234 (42.4)	240 (46.2)	237 (44.6)
Median [Min, Max]	231 [113, 696]	237 [129, 600]	234 [113, 696]
Missing	7 (0.4%)	45 (1.8%)	52 (1.2%)
BMI			
Mean (SD)	26.2 (3.40)	25.6 (4.52)	25.8 (4.08)
Median [Min, Max]	26.1 [15.5, 40.4]	24.8 [16.0, 56.8]	25.5 [15.5, 56.8]
Missing	4 (0.2%)	13 (0.5%)	17 (0.4%)

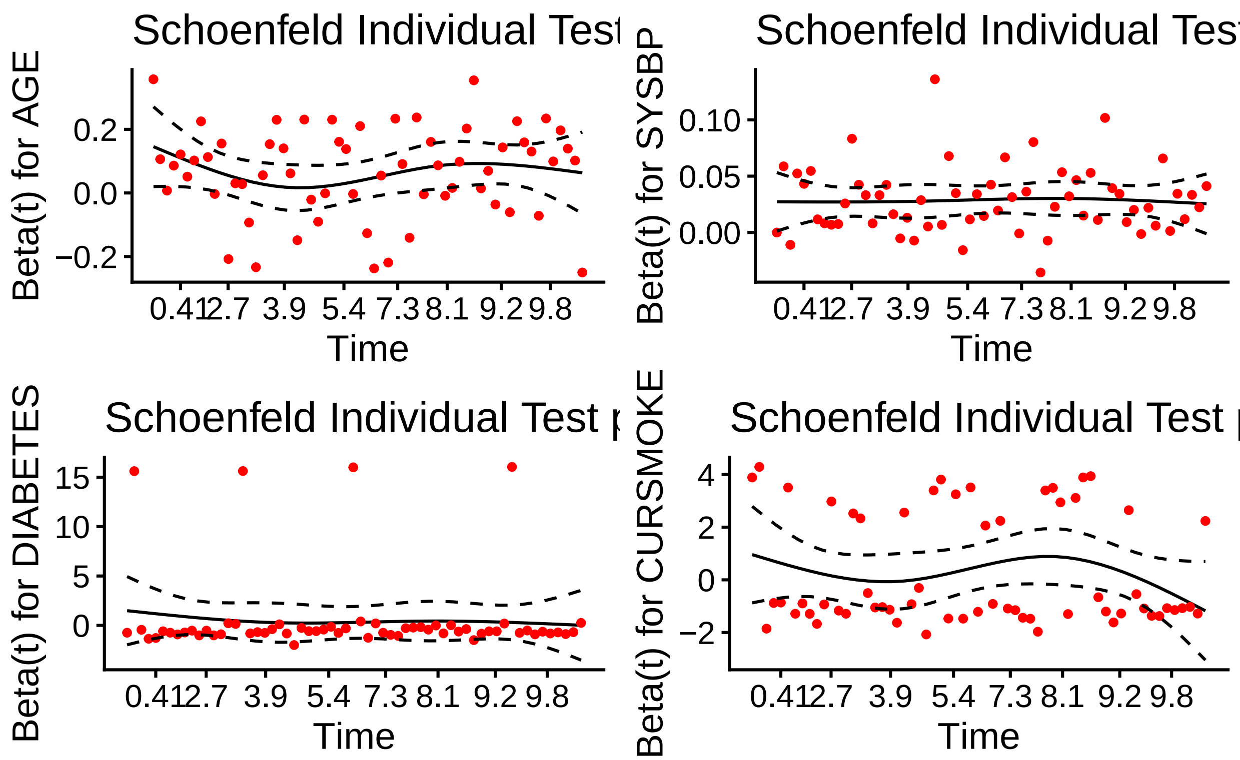


Stroke-Free Survival over 10 Years by Risk Factor
Survival (Females)
Survival (Males)
1.000
0.975
0.950
0.925
0.900
0
2.5
5
7.5
10
Years
Strata
DIABETES=No
DIABETES=Yes
Strata
HIGHBP=No
HIGHBP=Yes
Strata
CURSMOKE=No
CURSMOKE=Yes

Global Schoenfeld Test p: 0.2277

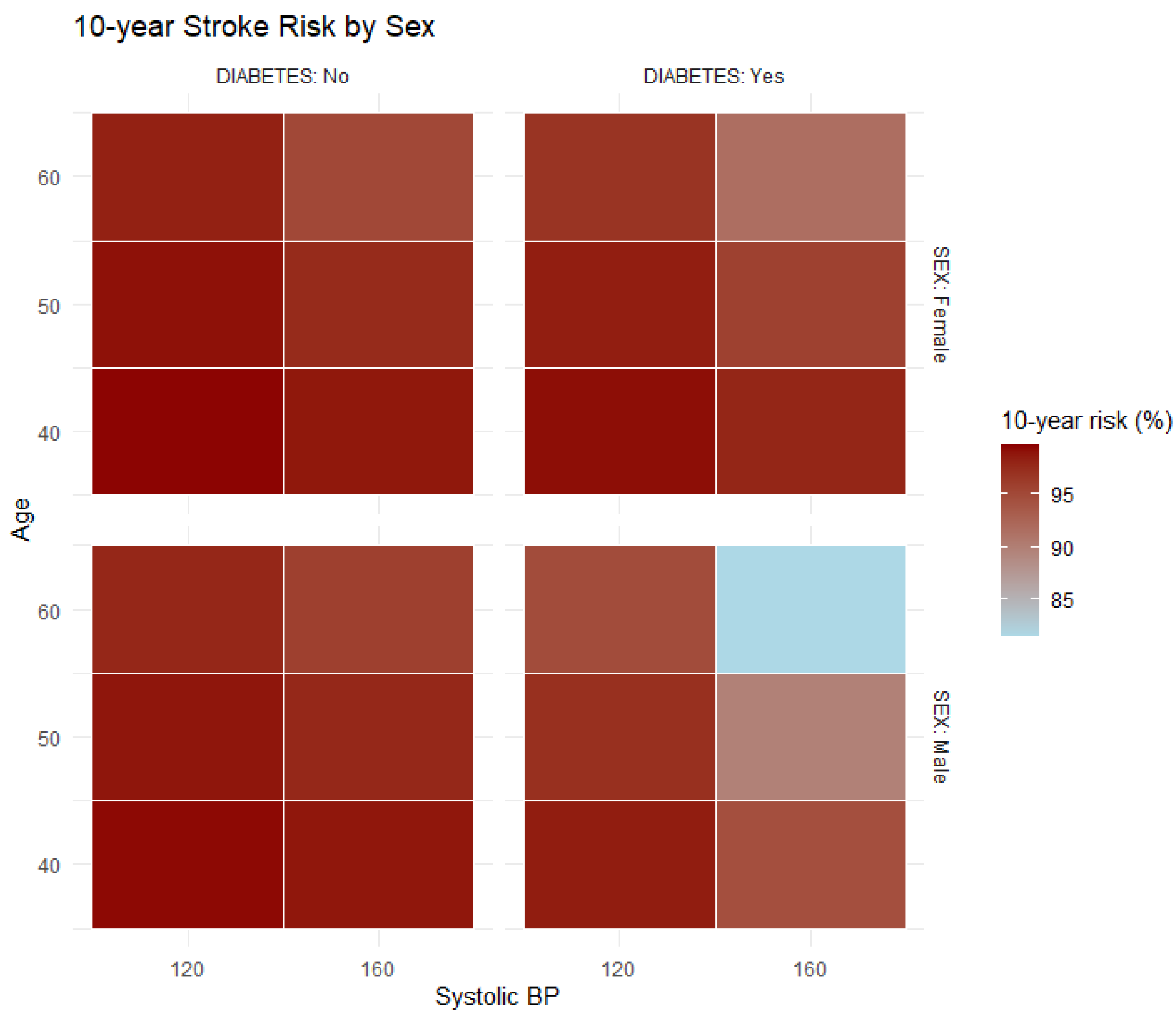


Global Schoenfeld Test p: 0.9575

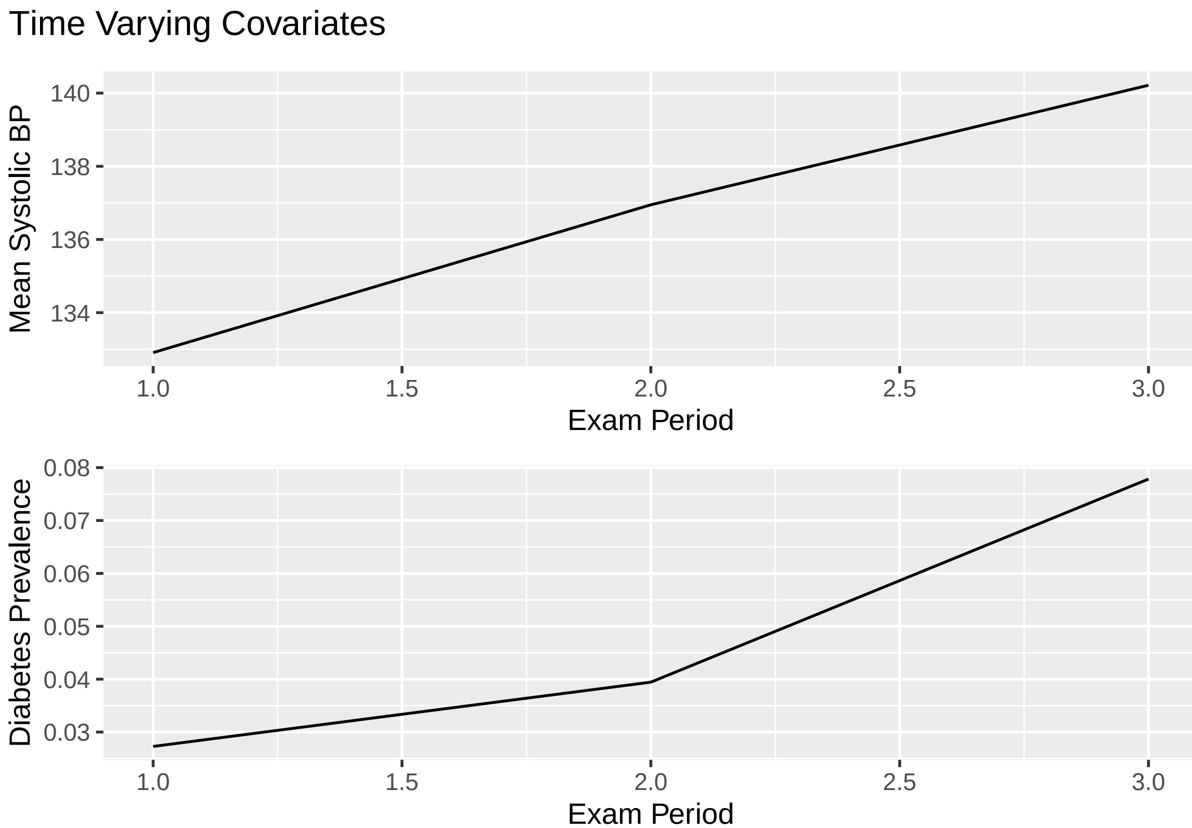


10-year Stroke Risk by Sex and Clinical Profile

	No Baseline Risk Factors	High Blood Pressure	Diabetes	High BP and Diabetes	Current Smoker
Female					
40.0	99.6	98.8	99.2	97.9	99.6
50.0	99.1	97.6	98.5	95.8	99.1
60.0	98.2	95.2	96.9	91.6	98.2
Male					
40.0	99.7	98.8	98.5	94.5	99.4
50.0	99.4	97.8	97.2	89.8	98.9
60.0	98.9	95.9	94.8	81.6	97.9



10-year Stroke Risk by Sex
DIABETES: No
DIABETES: Yes
SEX: Female
SEX: Male
Age
60
50
40
120
160
Systolic BP
10-year risk (%)
95
90
85


Time Varying Covariates
Mean Systolic BP
140
138
136
134
1.0
1.5
2.0
2.5
3.0
Exam Period
Diabetes Prevalence
0.08
0.07
0.06
0.05
0.04
0.03
1.0
1.5
2.0
2.5
3.0
Exam Period